

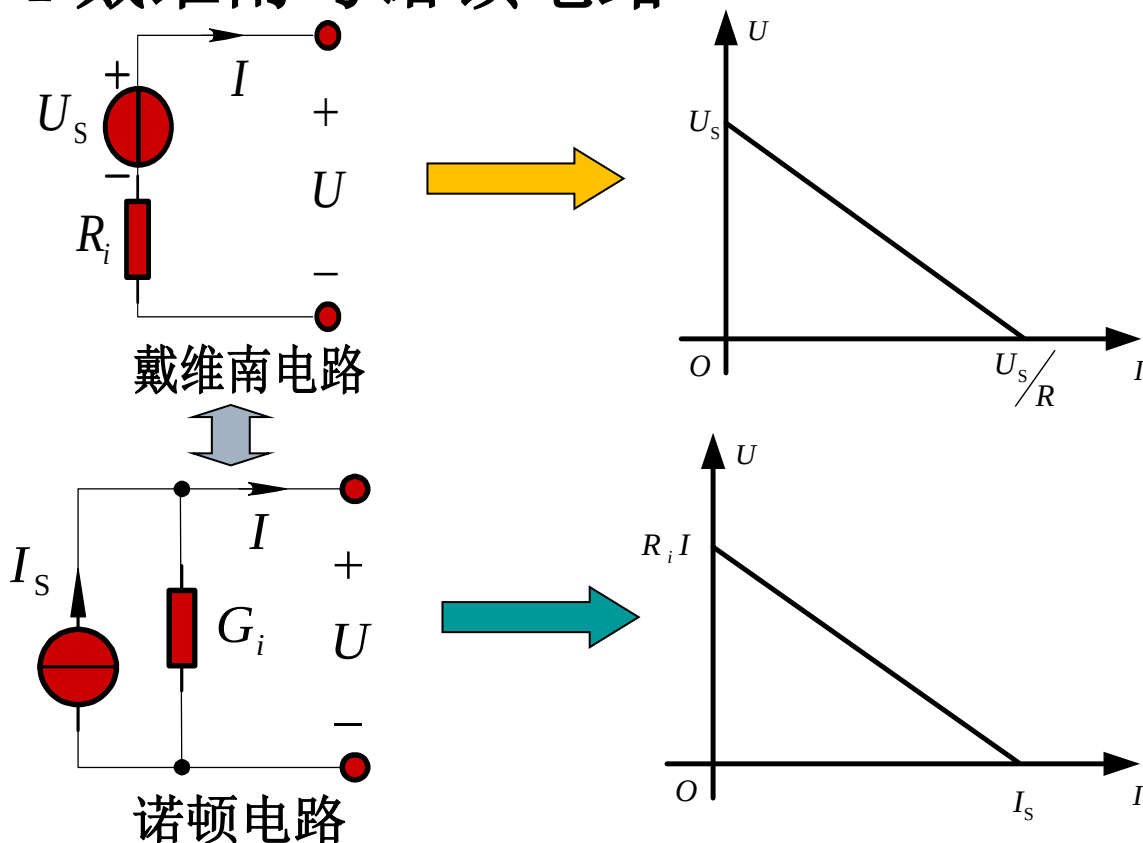
第二章 线性直流电路



2.2 含源支路的等效变换

基本要求：掌握各种含源支路的等效化简方法和戴维南、诺顿两种典型电路之间的等效变换规律，能熟练运用这些等效规律化简电路。

1 戴维南与诺顿电路



端口特性方程

$$U = U_S - R_i I$$

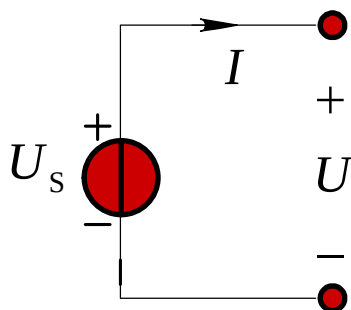
$$I = I_S - G_i U$$

等效条件为

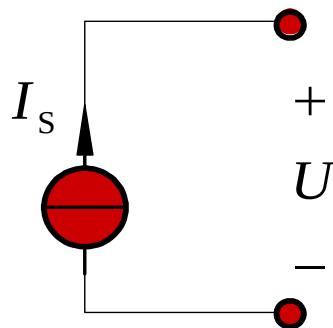
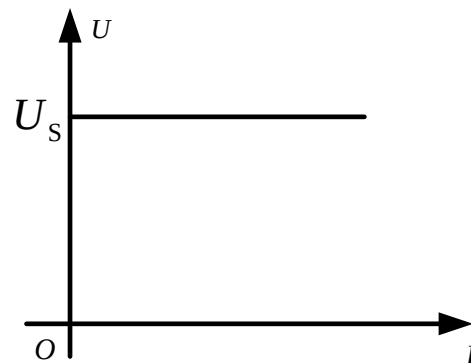
$$I_S = \frac{U_S}{R_i}, \quad G_i = \frac{1}{R_i}$$

2.2 含源支路的等效变换

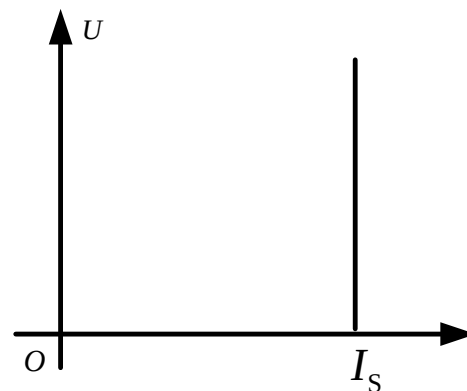
注：电压源内阻 $R_i=0$ ，而电流源内导 $G_i=0$ 时，即内阻等于无穷大，它们也称为理想电源。零不能取倒数，故电压源和电流源不能相互等效。



电压源



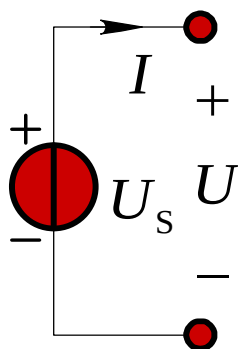
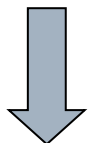
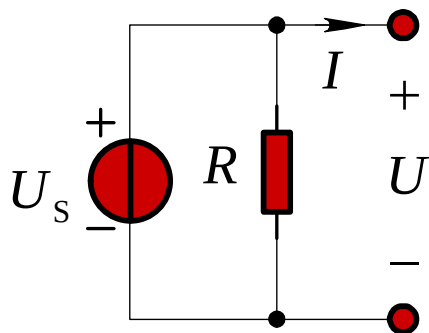
电流源



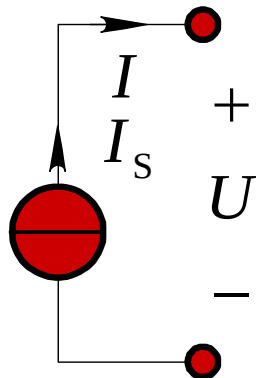
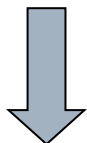
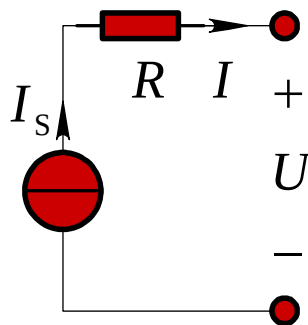
2.2 含源支路的等效变换

2 其它含源支路的等效

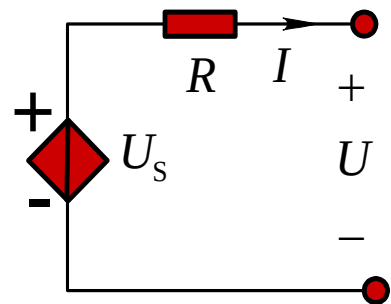
电压源并电阻



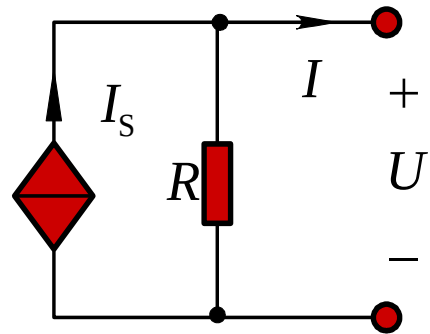
电流源串电阻



含受控源支路的等效



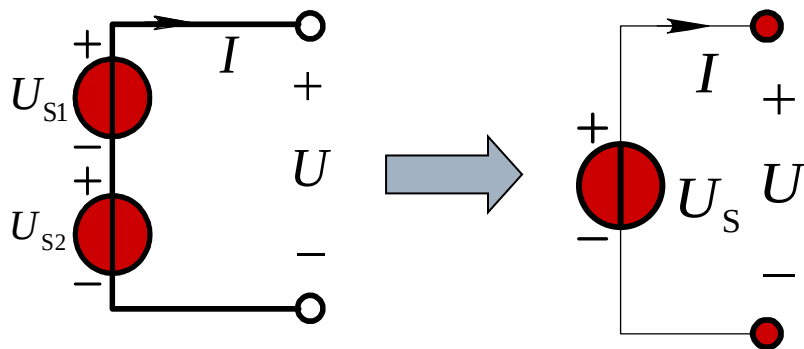
$$I_s = U_s / R \quad \longleftrightarrow \quad U_s = I_s R$$



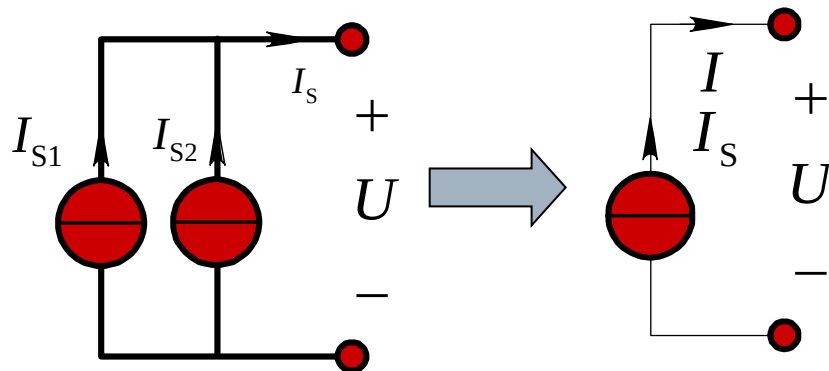
变换过程控制量位置不变

2.2 含源支路的等效变换

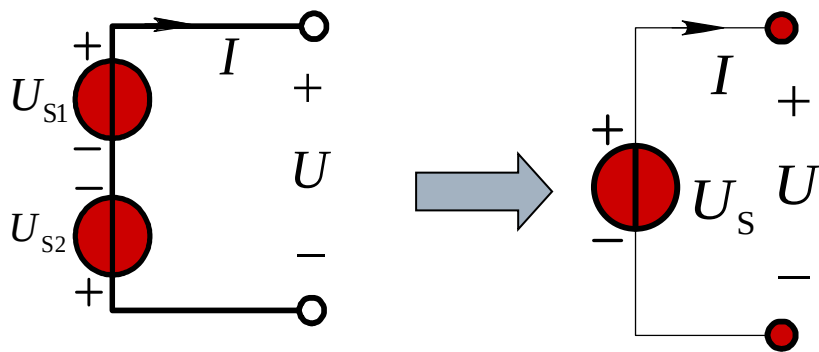
电压源的正向，反向串联 电流源的正向，反向并联



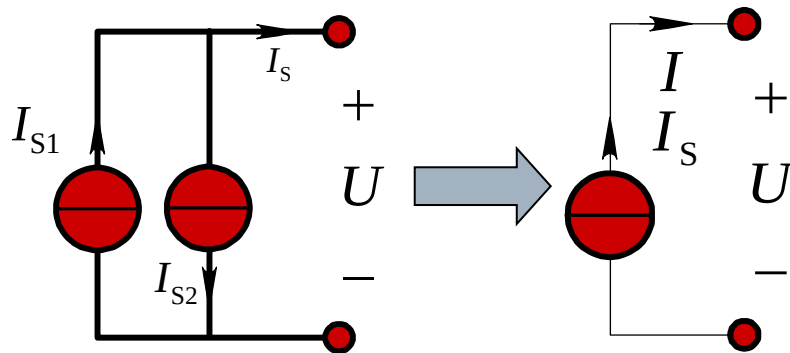
$$U_S = U_{S1} + U_{S2}$$



$$I_S = I_{S1} + I_{S2}$$



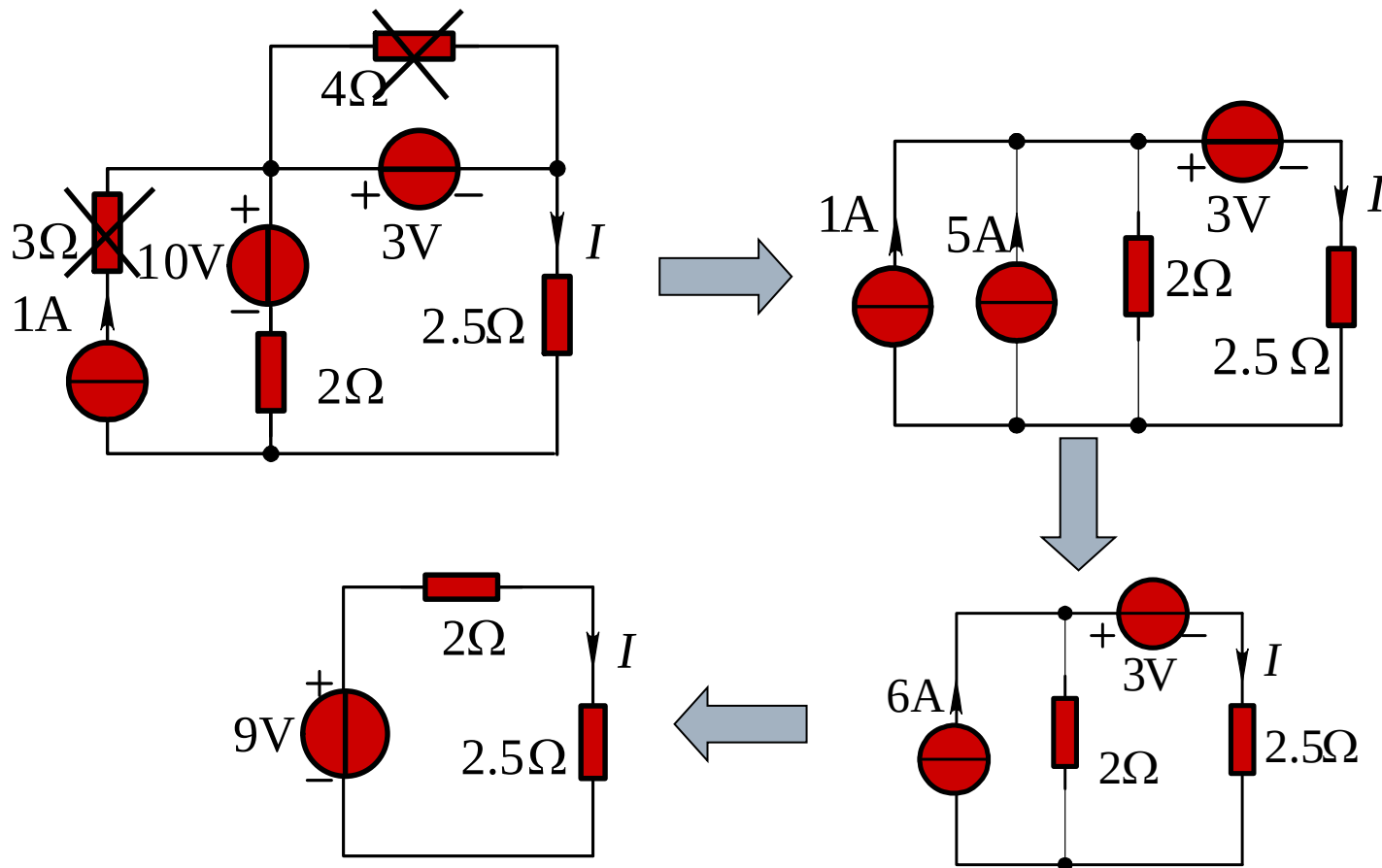
$$U_S = U_{S1} - U_{S2}$$



$$I_S = I_{S1} - I_{S2}$$

2.2 含源支路的等效变换

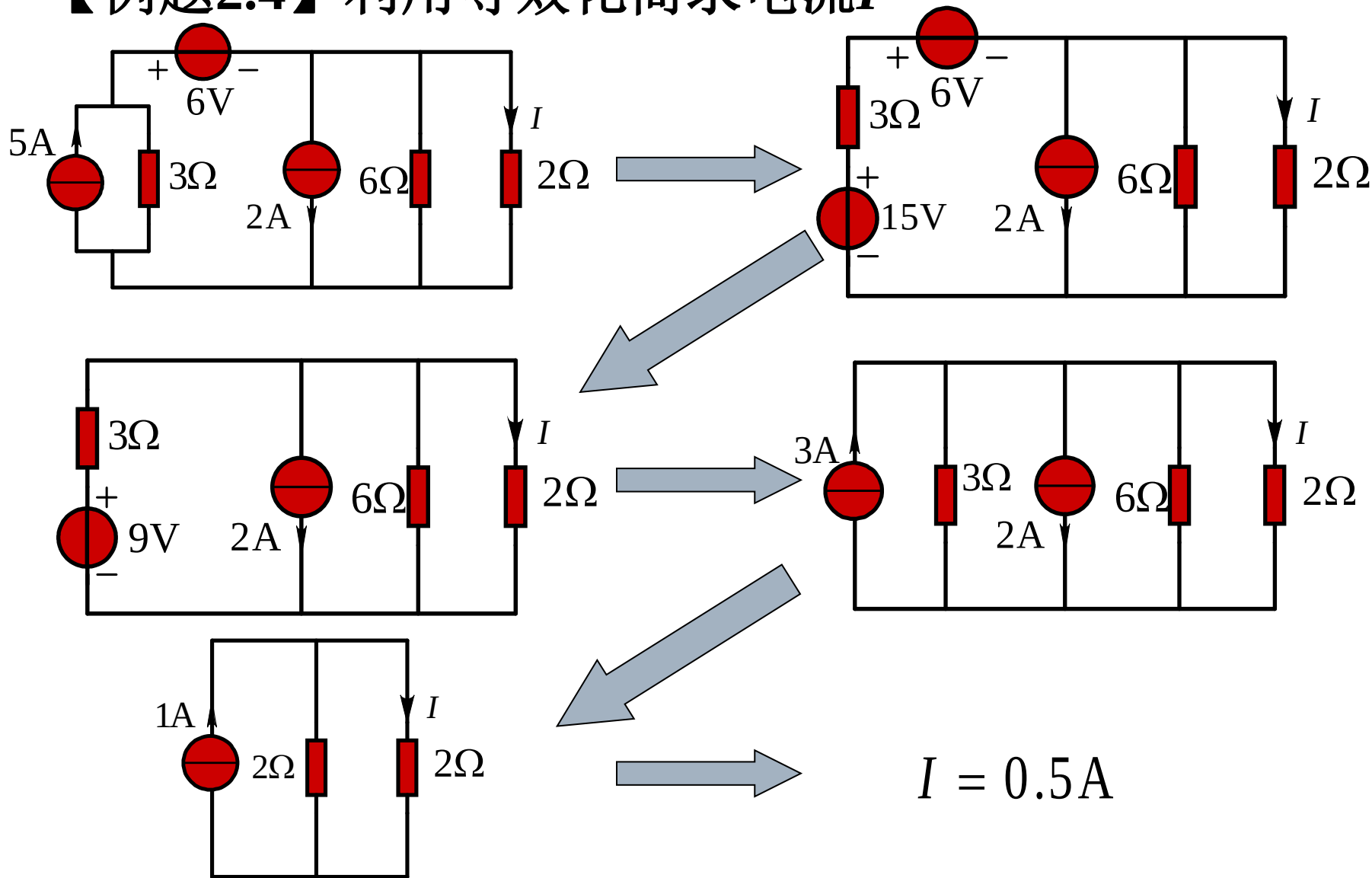
【例题2.3】用等效变换求图示电路中电流 I 。



$$I = \frac{9\text{V}}{2\Omega + 2.5\Omega} = 2\text{A}$$

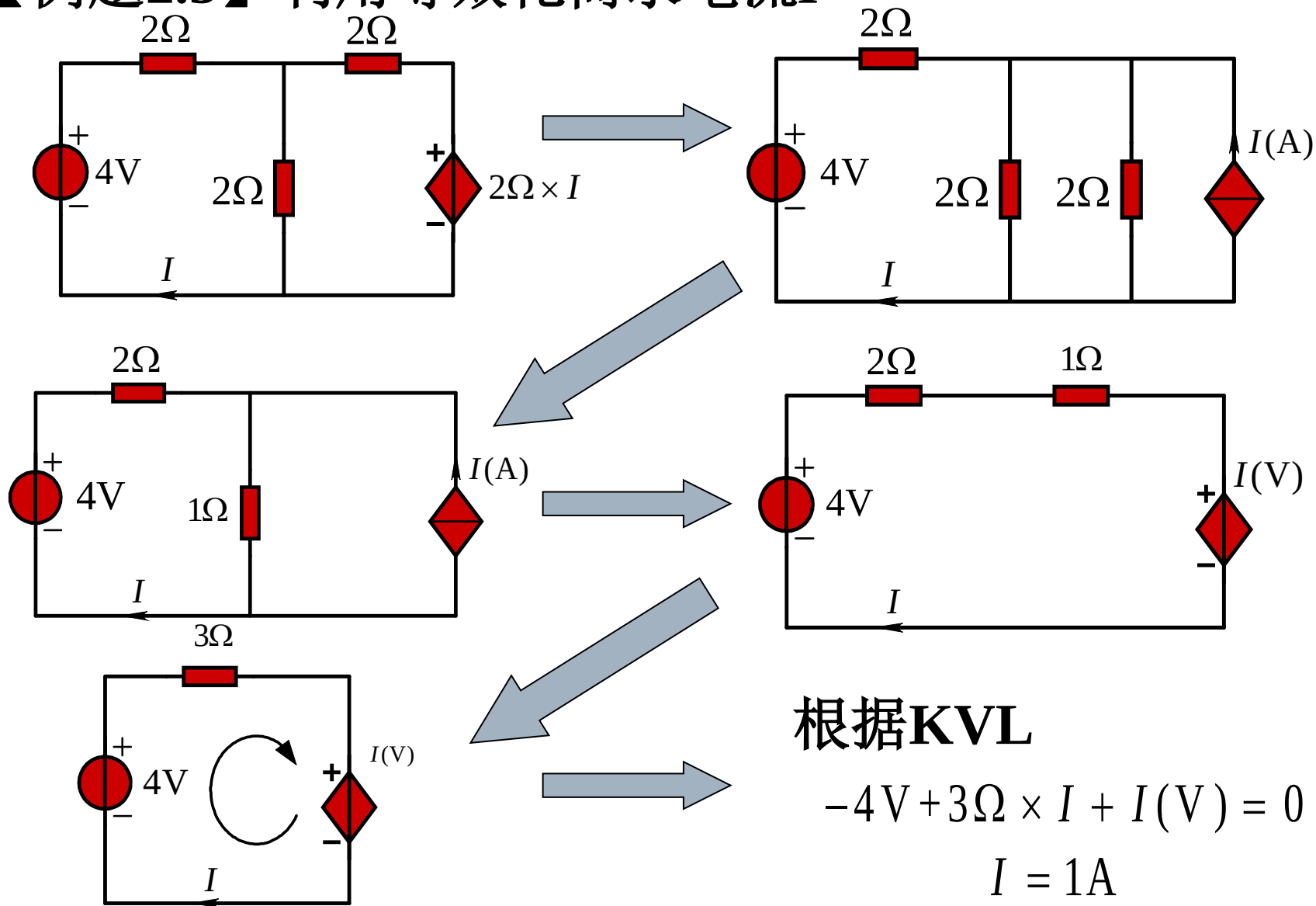
2.2 含源支路的等效变换

【例题2.4】利用等效化简求电流 I



2.2 含源支路的等效变换

【例题2.5】利用等效化简求电流 I



谢谢

谢谢！